5.7 POLYSACCHARIDES

Carbohydrates which contain more than 10 monosaccharide units are known as polysaccharides. Example : Starch, cellulose, glycogen, inulin etc.

5.7.1 HOMOPOLYSACCHARIDES

Starch:

a) Source:

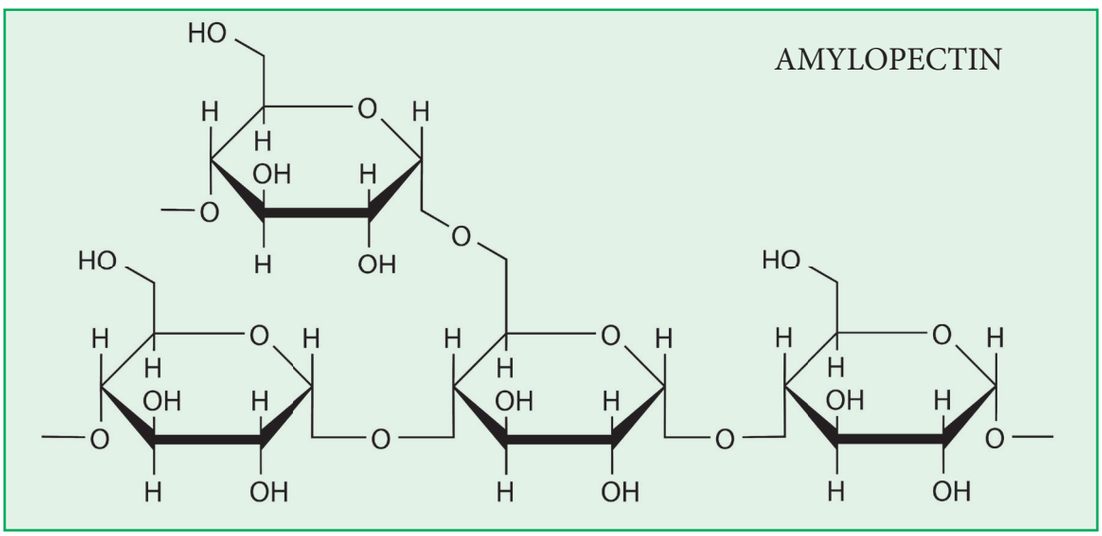
Plant materials such as roots, tubers, stem, vegetables, fruits and cereals are the main sources of starch.

b) Structure:

Starch is the nutritional reservoir in plants. Starch is a homopolysaccharide consists of only α-D-glucose. The two chief constituents of starch are (i) Amylose (15-20%) and (ii) Amylopectin (80 - 85%).

Amylose forms the inner portion of the starch grain and is soluble in water. It is a linear, non-branched polymer of glucose. The glucose residues are united by α(1-4) linkage. The molecular weight of amylose is 60,000.

Amylopectin forms the outer covering of the starch grain and is insoluble in water. It is a highly branched polymer of glucose. The glucose residues are united by α(1- 4) linkages in the chains and by α (1 - 6) at the branch points. Its molecular weight is 2,00,000. It is like glycogen except its lower degree of branching.



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